

Boss Challenge — Paper 01 (Class 7)

Heat | Nutrition in Plants | Integers

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. A clinical thermometer (for body temperature) usually has a scale of about:
(A) 0-100°C
(B) 35-42°C
(C) -10-110°C
(D) 0-50°C
2. Plants that make their own food using sunlight are called:
(A) heterotrophs
(B) autotrophs
(C) parasites
(D) saprotrophs
3. Find: $-5 + 8$
(A) -13
(B) 3
(C) 13
(D) -3
4. Heat always flows on its own from a:
(A) cold body to a hot body
(B) hot body to a cold body
(C) both ways equally
(D) it never flows
5. The tiny pores on a leaf through which gases enter and leave are called:
(A) chlorophyll
(B) stomata
(C) veins
(D) roots
6. Find: $(-4) \times (-6)$
(A) -24
(B) 24
(C) -10
(D) 10
7. We wear woollen clothes in winter mainly because wool:
(A) gives out heat
(B) traps air and stops body heat escaping
(C) is naturally cold
(D) absorbs the cold

8. In photosynthesis the gas taken IN and the gas given OUT are:
- (A) O₂ in, CO₂ out
 - (B) CO₂ in, O₂ out
 - (C) N₂ in, O₂ out
 - (D) CO₂ in, CO₂ out
9. The additive inverse (opposite) of -7 is:
- (A) -7
 - (B) 7
 - (C) 0
 - (D) 1/7
10. A sea breeze blows from sea to land in the daytime because:
- (A) the sea is hotter than land
 - (B) land heats faster, warm air over it rises, and cool air from the sea moves in
 - (C) wind comes from the mountains
 - (D) there is no real reason
11. Cuscuta (Amarbel), a leafless yellow climber, is a:
- (A) autotroph
 - (B) parasite
 - (C) insectivore
 - (D) saprotroph
12. Find: $12 \div (-3)$
- (A) 4
 - (B) -4
 - (C) -36
 - (D) 9
13. Two thin blankets keep you warmer than one thick blanket because:
- (A) two blankets are heavier
 - (B) air trapped between the two layers is a poor conductor (good insulator)
 - (C) thin blankets are made of warmer cloth
 - (D) there is no difference
14. The green pigment that traps sunlight for photosynthesis is:
- (A) starch
 - (B) chlorophyll
 - (C) glucose
 - (D) stomata
15. Which is greater: -3 or -7?
- (A) -7
 - (B) -3
 - (C) they are equal
 - (D) cannot say

16. Which type of heat transfer can happen even through empty space (a vacuum), like the Sun's heat reaching Earth?

- (A) conduction
- (B) convection
- (C) radiation
- (D) boiling

17. Insectivorous plants such as the pitcher plant trap insects mainly to get:

- (A) water
- (B) sunlight
- (C) nitrogen
- (D) oxygen

18. Find: $-15 + 15$

- (A) 30
- (B) 0
- (C) -30
- (D) 1

19. A metal spoon left in hot tea becomes hot at the far end by:

- (A) convection
- (B) radiation
- (C) conduction
- (D) evaporation

20. A few drops of iodine turn a leaf blue-black. This shows the leaf contains:

- (A) protein
- (B) starch
- (C) fat
- (D) only water

21. Find: $(-2) \times 3 \times (-1)$

- (A) -6
- (B) 6
- (C) 5
- (D) -5

22. In summer we prefer light-coloured clothes because light colours:

- (A) absorb more heat
- (B) reflect more heat, keeping us cooler
- (C) produce heat
- (D) make no difference

23. Plants and fungi that feed on dead, decaying matter are called:
- (A) autotrophs
 - (B) parasites
 - (C) saprotrophs
 - (D) insectivores
24. The temperature was 5°C and dropped by 8°C . The new temperature is:
- (A) 3°C
 - (B) -3°C
 - (C) 13°C
 - (D) -13°C
25. A clinical thermometer has a small kink (bend) near the bulb in order to:
- (A) look attractive
 - (B) stop the mercury slipping back so you can read it after removing it
 - (C) hold extra mercury
 - (D) cool the bulb
26. Rhizobium bacteria living in the root nodules of pulses (legumes) help the plant by:
- (A) giving it water
 - (B) fixing nitrogen from the air into a usable form
 - (C) eating insects
 - (D) making chlorophyll
27. Find: $0 \times (-99)$
- (A) -99
 - (B) 99
 - (C) 0
 - (D) 1
28. Convection currents (warm part rising, cool part sinking) are set up mainly in:
- (A) only solids
 - (B) liquids and gases
 - (C) a vacuum
 - (D) only metals
29. Lichen is a partnership (symbiosis) between:
- (A) two algae
 - (B) an alga and a fungus
 - (C) two fungi
 - (D) a plant and an insect
30. The statement $4 + (-4) = 0$ is an example of:
- (A) closure
 - (B) the additive inverse
 - (C) the commutative property
 - (D) the distributive property

31. A metal chair feels colder than a wooden chair at the same temperature because metal:

- (A) is actually colder
- (B) conducts heat away from your skin faster
- (C) is hot inside
- (D) there is no reason

32. The food made in the leaves is mainly stored as:

- (A) glucose only
- (B) starch
- (C) protein
- (D) vitamins

33. Find: $(-8) - (-3)$

- (A) -11
- (B) -5
- (C) 5
- (D) 11

34. At night a land breeze blows from land to sea because:

- (A) land is warmer at night
- (B) sea cools faster than land
- (C) the sea stays warmer, so cooler air from the land flows toward it
- (D) there is no reason

35. Each stoma (leaf pore) is opened and closed by a pair of:

- (A) veins
- (B) guard cells
- (C) chlorophyll grains
- (D) roots

36. Multiplying two integers with OPPOSITE signs gives a product that is:

- (A) positive
- (B) negative
- (C) zero
- (D) one

37. Which of these is the BEST conductor of heat?

- (A) wood
- (B) plastic
- (C) copper
- (D) wool

38. The energy used by a plant during photosynthesis comes from:

- (A) the soil
- (B) sunlight
- (C) the wind
- (D) water only

39. Find: $(-6) \times 0 + (-3)$

- (A) -3
- (B) -18
- (C) 3
- (D) 0

40. Heat is best described as a form of:

- (A) matter
- (B) energy
- (C) light only
- (D) a liquid

41. Farmers add fertilisers or manure to fields mainly to replace:

- (A) water
- (B) sunlight
- (C) nutrients such as nitrogen used up by crops
- (D) air

42. Arrange in increasing order: -2, 3, -5, 0

- (A) -2, -5, 0, 3
- (B) -5, -2, 0, 3
- (C) 3, 0, -2, -5
- (D) -5, 0, -2, 3

43. The shiny silver inner walls of a thermos (vacuum) flask reduce heat loss mainly by:

- (A) conduction
- (B) convection
- (C) radiation
- (D) boiling

44. The MAIN site of photosynthesis in most plants is the:

- (A) roots
- (B) leaves
- (C) flowers
- (D) bark of the stem

45. Today is -2°C . It rises 5°C , then falls 4°C . The final temperature is:

- (A) -1°C
- (B) 1°C
- (C) -11°C
- (D) 7°C

46. When water boils in a pot, the upper water gets hot mainly by:

- (A) conduction
- (B) convection currents
- (C) radiation
- (D) no transfer

47. An organism that cannot make its own food and depends on others is a:

- (A) autotroph
- (B) heterotroph
- (C) producer
- (D) chlorophyll

48. Find: $(-1) \times (-1) \times (-1) \times (-1)$

- (A) -1
- (B) 1
- (C) 4
- (D) -4

49. On a sunny day a black object and a white object are left outside. The black one gets hotter because dark colours:

- (A) reflect more heat
- (B) absorb more heat (radiation)
- (C) give out heat
- (D) stay cool

50. A pitcher plant is green and can photosynthesise, yet it still traps insects because its soil is poor in:

- (A) water
- (B) sunlight
- (C) nitrogen
- (D) carbon dioxide

51. The product $(-1) \times (-2) \times (-3) \times (-4) \times (-5)$ is:

- (A) positive
- (B) negative
- (C) zero
- (D) one

52. The normal temperature of a healthy human body is about:

- (A) 37°C
- (B) 42°C
- (C) 30°C
- (D) 50°C

53. Which of these takes food from a living host plant?

- (A) mushroom
- (B) Cuscuta (Amarbel)
- (C) pitcher plant
- (D) lichen

54. Which shows the COMMUTATIVE property of addition?

- (A) $-3 + 5 = 5 + (-3)$
- (B) $-3 + 0 = -3$
- (C) $-3 + 3 = 0$
- (D) $(-3+5)+2 = -3+(5+2)$

55. Cooking-pan handles are made of plastic or wood because these materials:

- (A) look nice only
- (B) are insulators, so the handle stays cool enough to hold
- (C) conduct heat very fast
- (D) are cheap only

56. The gas released by plants in photosynthesis that is useful for us to breathe is:

- (A) carbon dioxide
- (B) oxygen
- (C) nitrogen
- (D) water vapour

57. Find: $(-9) + (-6)$

- (A) -15
- (B) -3
- (C) 15
- (D) 3

58. Heat travels through empty space from the Sun to the Earth by:

- (A) conduction
- (B) convection
- (C) radiation
- (D) none of these

59. Sunlight, chlorophyll, carbon dioxide and water are needed for photosynthesis. Which one is taken from the SOIL?

- (A) sunlight
- (B) carbon dioxide
- (C) water
- (D) chlorophyll

60. The value of $|-8|$ (its distance from 0 on the number line) is:

- (A) -8
- (B) 8
- (C) 0
- (D) 16

61. A laboratory thermometer is built to measure a wide range, roughly:
- (A) 35-42°C
 - (B) -10°C to 110°C
 - (C) 0-50°C
 - (D) 90-110°C
62. In a food chain, green plants are called ____ because they make food.
- (A) consumers
 - (B) producers
 - (C) decomposers
 - (D) parasites
63. Find: $(-20) \div (-5) - 3$
- (A) 1
 - (B) -7
 - (C) 7
 - (D) -1
64. Sea breeze and land breeze are everyday examples of heat transfer in air by:
- (A) conduction
 - (B) convection
 - (C) radiation
 - (D) none
65. Mushrooms growing on a rotting log get their food by:
- (A) photosynthesis
 - (B) trapping insects
 - (C) absorbing nutrients from dead and decaying matter
 - (D) directly from sunlight
66. Which number is the additive identity (adding it changes nothing)?
- (A) 1
 - (B) 0
 - (C) -1
 - (D) 10
67. Holding a metal rod in a flame, the far end gets hot. The metal's particles:
- (A) move bodily to the far end
 - (B) stay in place but vibrate and pass energy to their neighbours
 - (C) disappear
 - (D) cool down
68. Which statement about photosynthesis is FALSE?
- (A) it needs sunlight
 - (B) it releases oxygen
 - (C) it happens in the green parts of a plant
 - (D) it happens mainly at night

69. Fill in the blank: $(-7) \times \blacksquare = 0$. The value of $\blacksquare$ is:

- (A) -7
- (B) 7
- (C) 0
- (D) 1

70. Which is the best choice of clothing for a hot, sunny afternoon?

- (A) thick black woollens
- (B) light-coloured, loose cotton
- (C) dark, tight nylon
- (D) two sweaters

71. Air has plenty of nitrogen gas, but most plants cannot use it directly. It is made usable by:

- (A) sunlight
- (B) Rhizobium bacteria (nitrogen fixation)
- (C) stomata
- (D) chlorophyll

72. Find: $(-3) - (-3) - (-3)$

- (A) 3
- (B) -9
- (C) -3
- (D) 9

73. Why does warm air rise?

- (A) warm air is heavier
- (B) warm air is lighter (less dense) than cool air
- (C) warm air becomes solid
- (D) there is no reason

74. Stomata usually close at night to:

- (A) take in more light
- (B) reduce water loss when photosynthesis isn't happening
- (C) release starch
- (D) absorb soil water

75. On a number line, starting at -4 and moving 6 steps to the RIGHT lands you on:

- (A) 2
- (B) -10
- (C) 10
- (D) -2

76. Two vessels of water, one shiny silver and one dull black, are left in the sun. Which heats up faster?
- (A) the silver one
 - (B) the black one
 - (C) both the same
 - (D) neither heats
77. Which raw material for photosynthesis comes from the AIR?
- (A) water
 - (B) carbon dioxide
 - (C) minerals
 - (D) chlorophyll
78. $(-5) + (-5) + (-5)$ is the same as:
- (A) $(-5) \times 3 = -15$
 - (B) $(-5) - 3$
 - (C) 5×3
 - (D) $(-5) \div 3$
79. Heat moves through a solid block of iron mainly by:
- (A) conduction
 - (B) convection
 - (C) radiation
 - (D) evaporation
80. Amarbel (*Cuscuta*) is pale yellow with no green colour. This tells us it:
- (A) makes plenty of its own food
 - (B) cannot photosynthesise, so it lives as a parasite
 - (C) is a saprotroph
 - (D) traps insects
81. The sum of two negative integers is always:
- (A) positive
 - (B) negative
 - (C) zero
 - (D) one
82. A woollen blanket keeps us warm because wool:
- (A) produces heat
 - (B) is a poor conductor and traps warm air
 - (C) is a good conductor
 - (D) reflects light

83. In bright daylight a green plant gives out more of which gas overall?
- (A) carbon dioxide
 - (B) oxygen
 - (C) nitrogen
 - (D) hydrogen
84. Which is correct?
- (A) $(-2) + (-3) = -5$
 - (B) $(-2) + (-3) = 5$
 - (C) $(-2) + (-3) = 1$
 - (D) $(-2) + (-3) = -1$
85. The cool daytime breeze that helps fishermen near the coast is caused by:
- (A) radiation alone
 - (B) uneven heating of land and sea, which sets up convection
 - (C) conduction through the water
 - (D) the moon
86. Which of these is NOT needed for photosynthesis?
- (A) sunlight
 - (B) carbon dioxide
 - (C) oxygen
 - (D) water
87. Evaluate: $6 + (-4) \times 2$
- (A) 4
 - (B) -2
 - (C) 20
 - (D) 8
88. Which statement is correct?
- (A) heat and temperature are the same thing
 - (B) temperature tells how hot a body is; heat is the energy that flows
 - (C) heat is measured in $^{\circ}\text{C}$
 - (D) temperature flows from hot to cold
89. Leaves are usually broad and flat mainly so they can:
- (A) look attractive
 - (B) catch the maximum sunlight for photosynthesis
 - (C) store water
 - (D) trap insects
90. A diver goes 8 m below sea level (-8 m), then rises 3 m. Her new position is:
- (A) -5 m
 - (B) -11 m
 - (C) 5 m
 - (D) 11 m

91. Hot food in a steel box stays warm longer if wrapped in a woollen cloth because the cloth:

- (A) adds heat to the food
- (B) slows heat loss by trapping air (insulation)
- (C) cools the food faster
- (D) conducts heat out quickly

92. The iodine starch test is done after keeping a plant first in the dark, then in light, to prove that ____ is needed for photosynthesis.

- (A) water
- (B) sunlight
- (C) soil
- (D) air

93. Which is the SMALLEST integer: -1, -10, 5, 0?

- (A) -1
- (B) -10
- (C) 0
- (D) 5

94. Heat energy is measured in joules. The instrument that measures temperature is a:

- (A) barometer
- (B) thermometer
- (C) ammeter
- (D) ruler

95. Which correctly shows the flow of photosynthesis?

- (A) glucose + O₂ → CO₂ + water
- (B) carbon dioxide + water → (sunlight, chlorophyll) → glucose + oxygen
- (C) oxygen + water → glucose
- (D) carbon dioxide + oxygen → water

96. Evaluate: $(-24) \div 6 \div (-2)$

- (A) 2
- (B) -2
- (C) 8
- (D) -8

97. A metal spoon, a wooden spoon and a plastic spoon are placed in hot soup. After a minute, whose handle feels hottest?

- (A) the wooden one
- (B) the plastic one
- (C) the metal one
- (D) all equally

98. Which of these is an autotroph (makes its own food)?

- (A) a mushroom
- (B) green grass
- (C) a human being
- (D) Cuscuta (Amarbel)

99. Evaluate: $(-10) \div 5$

- (A) -2
- (B) 2
- (C) -15
- (D) 15

100. Photosynthesis takes place mainly in the plant cells that contain:

- (A) chlorophyll
- (B) stored starch grains
- (C) trapped air
- (D) only water